

1. Business Understanding:

Unfortunately, even with several years of conservation projects aimed at breeding abalone, they are still facing extinction, and by all indications, extinction is inevitable. Therefore, there is a need for a model that can predict relevant information related to their size and other characteristics, whether for commercial purposes or for conservation. With a model that works correctly, we can better manage and monitor the population and, perhaps, save them from extinction.

2. Data Understanding:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4180 entries, 0 to 4179
Data columns (total 9 columns):
 #   Column                Non-Null Count  Dtype  
---  -
 0   Sex                   4180 non-null   object  
 1   Length                4180 non-null   float64  
 2   Diameter              4180 non-null   float64  
 3   Height                4172 non-null   float64  
 4   Whole_weight          4180 non-null   float64  
 5   Shucked_weight        4175 non-null   float64  
 6   Viscera_weight        4175 non-null   float64  
 7   Shell_weight          4180 non-null   float64  
 8   Rings                 4180 non-null   int64  
dtypes: float64(7), int64(1), object(1)
memory usage: 294.0+ KB
```

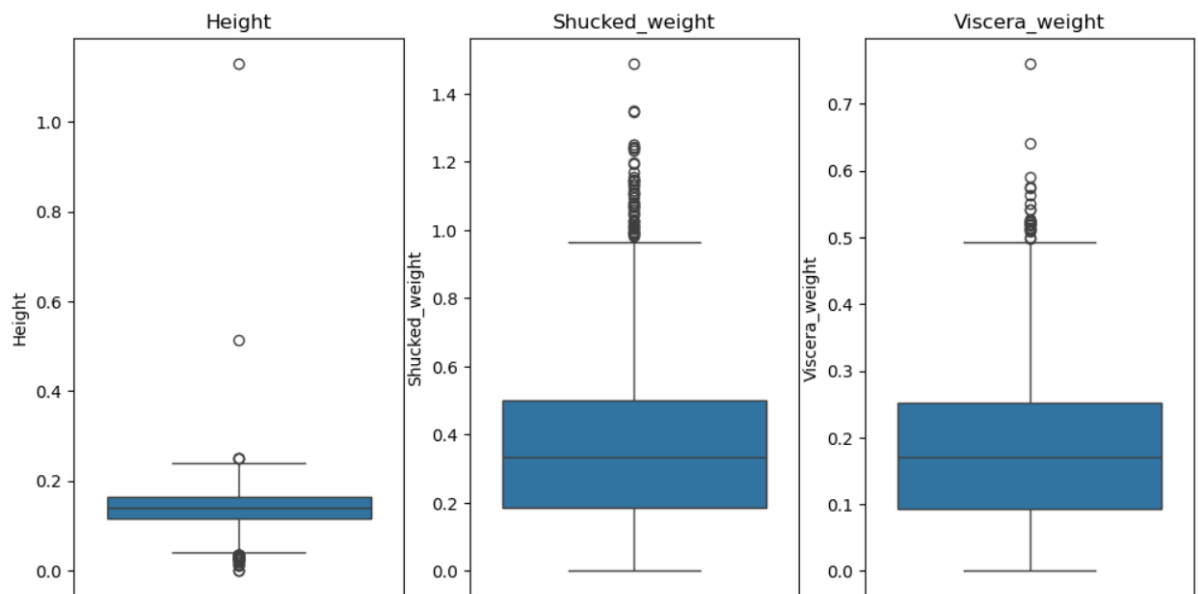
The collected database consists of 4180 rows and 9 columns, with very few null values. Overall, it's a dataset with limited resources, but it's accessible to all users, not requiring resource engineering.

	Length	Diameter	Height	Whole_weight	Shucked_weight	Viscera_weight	Shell_weight	Rings
count	4172.000000	4172.000000	4172.000000	4172.000000	4172.000000	4172.000000	4172.000000	4172.000000
mean	0.524041	0.407910	0.139529	0.828988	0.359474	0.359474	0.238906	9.932407
std	0.120140	0.099276	0.041804	0.490535	0.222032	0.222032	0.139250	3.224551
min	0.075000	0.055000	0.000000	0.002000	0.001000	0.001000	0.001500	1.000000
25%	0.450000	0.350000	0.115000	0.441500	0.186375	0.186375	0.130000	8.000000
50%	0.545000	0.425000	0.140000	0.799750	0.336000	0.336000	0.234000	9.000000
75%	0.615000	0.480000	0.165000	1.154000	0.502250	0.502250	0.329125	11.000000
max	0.815000	0.650000	1.130000	2.825500	1.488000	1.488000	1.005000	29.000000

The summary statistic above confirms a dataset that fluctuates with very high and low values. This is explained by the fact that the Sex feature has 3 classes (M, F, I), therefore it's natural for an adult abalone to be M (Male) or F (Female), which is much greater than I (Infant). However, care was taken considering the information about the dataset, some characteristics such as units MM (Millimeters), G (Grams), and Count (Float). There are also 3 rows of duplicate data that will be removed.

3. Data Preparation:

To prepare the dataset for training, the null values were replaced with the median, considering, as seen in plot:



the large variation in values, which is why the median was more suitable. The boxplots of Height, Shucked_weight and Viscera_weight indicated the presence of outliers, validating the use of the median as it is more stable than the mean, to replace these null values. LabelEncoder was chosen over OneHotEncoder to avoid adding extra columns, and over OrdinalEncoder to avoid implying a hierarchy between M, F and I. Three duplicate rows were detected and excluded. Of the 9 columns, 8 excluding the target feature, the columns indicating the

weight of the abalone viscera and meat were not numeric, preventing their use in the algorithm. Therefore, both features were converted from categorical to numeric. Only the target feature remained, which was categorical, but within a for loop, all columns that were still categorical were converted into numbers after the encoding process was applied, thus creating numeric columns. As mentioned before, due to different units of measurement and the large variation in values, the variable X was normalized. Finally, the distribution among the values of the target variable was checked to understand if there is a balance between them.

4. Modelling:

With the data already divided between training (90%) and testing (10%), the data is truly ready to be trained. To compare performance with different models, the data was trained using the K-Nearest Neighbours, Decision Tree Classifier, Logistic Regression, and Random Forest Classifier algorithms. Considering that the result found was not satisfactory and to meet the project requirements, another feature was tested as a target. To find this feature, the correlation matrix was useful for understanding the clarity between the features. As a result, the feature "Shucked_weight" was selected as the target, given that it showed a strong relationship with all other features. Furthermore, the data were separated into training (90%) and test (10%) sets and trained on the Linear Regression, Lasso Regression, and Ridge Regression models. Using Pipeline, it was possible to train these 3 models, with 2 of them being regularized.

5. Evaluation:

With the data already divided between training (90%) and testing (10%), the data is truly ready to be trained. To compare performance with different models, the data was trained using the K-Nearest Neighbour, Decision Tree Classifier, Logistic Regression, and Random Forest Classifier algorithms.

Initially, these values were obtained:

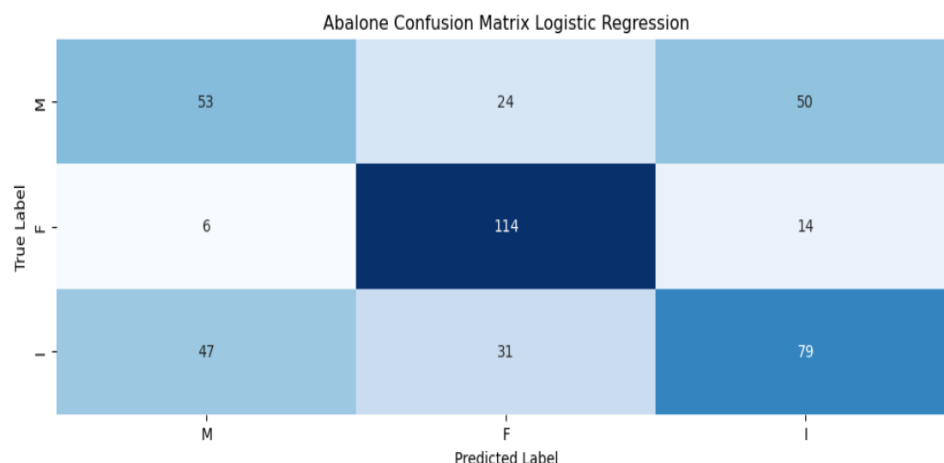
----- Model performance on test sample -----					
	Model	Accuracy	Precision	Recall	F1-Score
0	K-Nearest Neighbors	0.531100	0.527233	0.531100	0.525967
1	Decision Tree	0.492823	0.490018	0.492823	0.490718
2	Logistic Regression	0.588517	0.575657	0.588517	0.577261
3	Random Forest	0.576555	0.568024	0.576555	0.571237

The accuracy values correspond to the percentage of correct predictions of the entire dataset. Precision corresponds to the correct predictions of the positive class. Recall is the percentage of true positives that were

correctly identified by the model. Finally, the F1-score is a balance between recall and precision, thus proving the equilibrium between the two metrics. Cross-validation was also performed to determine if there was overfitting (the model being very well trained, it may have memorized too much, so that when new data is entered, the result is very different) or underfitting (still in training, the model cannot obtain good accuracy in training, obviously reflecting in low performance). **Logistic Regression achieved the highest mean CV accuracy (55.7%), confirming greater generalisation across folds.**

Model Performance using cross validation		
	Model	Mean CV Accuracy
0	K-Nearest Neighbors	0.524056
1	Decision Tree	0.496048
2	Logistic Regression	0.553272
3	Random Forest	0.538186

Another evaluation metric used was the Confusion Matrix:



In which it is visible that the model is far behind, considering that in 2 of the 3 classes, it sees more errors than correct answers.

Considering the results obtained above and the project requirements, with "Shucked weight" as the target characteristic, within the created pipeline, and with new models used, we obtained the following metrics in each of the models.

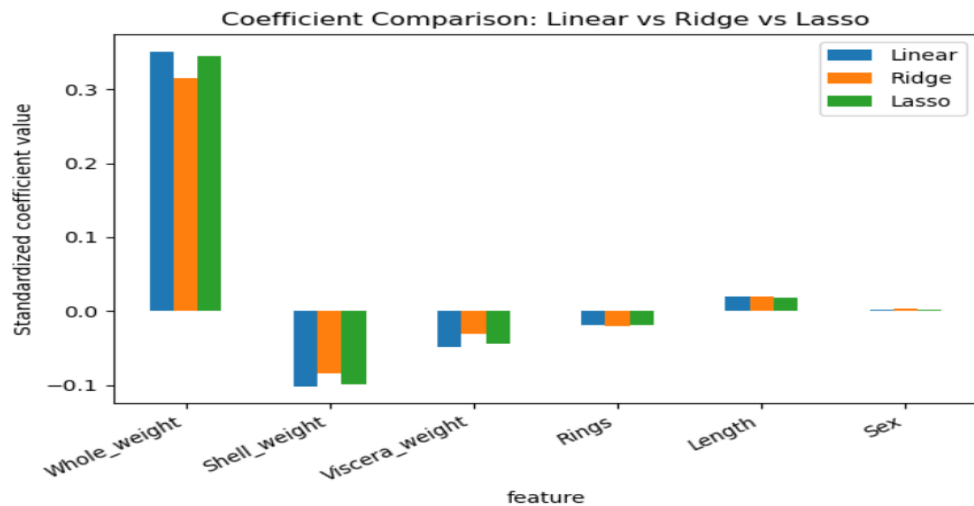
Test Performance (MSE, R^2)				
Linear		MSE = 0.0019		MAE = 0.0237 R^2 = 0.9546
Ridge		MSE = 0.0018		MAE = 0.0244 R^2 = 0.9574
Lasso		MSE = 0.0019		MAE = 0.0238 R^2 = 0.9552

In these 3 evaluated metrics, MSE and MAE are metrics that penalize errors in different ways, and it is up to the analyst to decide which one is more important or any relationship of that type. The main difference between them is that MSE is quadratic, so the errors weigh more, and make the analysis a little more complicated to interpret, compared to MAE, which is more robust and easier to interpret. As we can see, both are low, so there is no need to evaluate whether we should focus on one of these metrics. R2, on the other hand, generates the coefficient of variation of the predictions, in other words, the variations that it can explain, so if it explains, it predicts correctly.

Great results, however, it is important to understand the importance of each variable, calculating their coefficients.

```
Coefficients (sorted by |Linear|)
      feature   Linear   Ridge   Lasso
Whole_weight  0.351338  0.315760  0.345045
Shell_weight -0.101984 -0.085001 -0.098587
Viscera_weight -0.048383 -0.031527 -0.044542
      Rings -0.019274 -0.021093 -0.019395
      Length 0.018974  0.020357  0.018817
      Sex    0.002309  0.002766  0.002283
      Diameter 0.001979  0.004164  0.001673
      Height 0.000778  0.000655  0.000444

Lasso set 0 out of 8 coefficients to (near) zero.
```



In both images, we can see the result obtained numerically and visually. It is worth highlighting the importance of looking for a coefficient = 0 in the Lasso Regression model. This 0 would represent a variable without importance. Although a negative number is less than zero, negative coefficients are important because they have a negative relationship with the target feature. With all this information and having, in principle, a good model, the 10-fold cross-validation returned an average R^2 of 0.967, combined with a low standard deviation, confirming the model's consistency and the absence of overfitting, as this validation eliminates the "luck" factor in the data split. This is further explained by the high multicollinearity in the dataset (Length–Diameter: 0.99), where Ridge's coefficient shrinkage improves generalisation. Although it cannot be implemented in production, the result allows for precise monitoring to ensure a balance between the survival of the species and, in a way, that those who do it receive a financial return for it.

6. Deployment:

This step does not apply to the job, given that it is not possible to put this model into production.

7. REFERENCES

Decision Tree pdf theory - Pdf file from week 2(12-18 Jan)

ML_PRACTICE - KNN .ipynb file from week 3(19-25 Jan)

Ridge and Lasso Regression pdf - file from week 5

Linear-Rdige-Lasso .ipynb file from week 5

Comparison-Classification-Model -.ipynb file from week 5.2 (Saturday class)

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